W-B-30A
October 29, 1976
SUPERSEDING
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March 27, 1956

FEDERAL SPECIFICATION

BALLAST, FLUORESCENT LAMP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers ballasts which have rated open-circuit voltages of 2000 volts or less and are intended to operate at a frequency of 50 Hz or 60 Hz and ambient temperature of 50°F to 105°F. The above includes ballasts for use with hot-cathode fluorescent lamps of the switch-start (preheat-start), rapid-start (continuously heated cathodes) or instant-start type; and ballasts for use with cold-cathode fluorescent lamps. Ballasts covered herein shall conform to ANSI Standard C 82.1.

1.2 Classification. Ballasts furnished under this specification shall be either "Class P (UL Class P)" or "Other Than Class P" for use with the above indicated hot or cold cathode fluorescent lamps. Ballast Class (either "Class P" or "Other than Class P"), type (preheat-starting, instant starting, rapid starting or cold cathode), voltage rating, frequency, and capacity (number of lamps and wattage or current of each lamp) shall be specified by the contracting officer (see 6.1). Refer to ANSI C82.1 as necessary for discussion of ballast types, voltage ratings, and capacities.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:

MIL-P-116 - Preservation - Packaging, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-463 - Definitions and Systems of Units Electromagnetic Interference Technology.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute, Inc. (ANSI) Standards:

C82.1 - Specifications for Fluorescent Lamp Ballasts.
C82.2 - Methods of Measurement of Fluorescent Lamp Ballasts.
C82.3 - Specifications for Fluorescent Lamp Reference Ballasts.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

Certified Ballast Manufacturers.

Procedural Guide Certified Ballast Manufacturers.

(Application for copies should be addressed to the Certified Ballast Manufacturers, 2120 Keith Building, Cleveland, OH 44115.)

Underwriters' Laboratories (UL), Inc. Standard:

No. 935 - Fluorescent Lamp Ballasts.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Certified Ballast Manufacturers Certification and alternative thereto. Ballasts furnished under this specification shall be a product that has been "Certified" by the Certified Ballast Manufacturers (CBM) and is listed in the current CBM listed Products Directory. The CBM seal affixed to ballast shall be accepted as evidence of certification. As an alternative to CBM certification, those manufacturers (suppliers) not participating in the CBM program shall provide evidence that their product meets the requirements of ANSI Standard C82.1. Evidence of meeting ANSI C82.1 requirements shall be in the form of e

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"Certified Test Report", dated no earlier than one year prior to invitation for bids, indicating C82.1 compliance, from an independent test laboratory acceptable to the Government. All costs incurred in either "CBM certification" or independent laboratory testing shall be borne by manufacturer or supplier. Evidence of "CBM Certification" or "Certified Test Report" shall be presented to the Government prior to contract award, at a time indicated by the Government's representative.

3.2 Fire and casualty hazards. Each contractor shall submit proof that the ballasts he proposes to supply under this specification conform to the requirements of UL Standard No. 935. Evidence of conformance with UL Standard No. 935 shall be the Underwriters' Laboratories Listing Mark affixed to ballast. In lieu of the Underwriters' Listing, Mark, the contractor may submit a test report from an independent laboratory acceptable to the Government, indicating that ballasts conform to applicable requirements of the published standard, including method of tests of UL Standard No. 935. Compliance with the above preliminary requirements in regard to fire, safety and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure acceptance of his material or equipment.

3.3 Material. All material used in construction of ballasts shall be new and suitable for the purpose intended. Also, material shall be free of defects that affect ballast appearance, serviceability, or safety. Metals shall be protected against corrosion. Refer to 4.4.1 for visual examination.

3.4 Dimensions. Unless otherwise specified, (see 6.1) "brick type " ballasts shall be furnished. The maximum over-all dimensions of 40 and 100 watt, two lamp brick type ballasts shall not exceed the following:

Capacity	Maximum dimensions (inches)
Two - 40 watt lamps	2-5/8 x 3-5/16 x 9-5/8
Two - 100 watt lamps	2-13/16 x 3-3/16 x 19-1/2

NOTE: The maximum over-all dimensions of other capacity and shape ballasts shall be specified (see 6.1). Refer to 4.4.1 for visual examination.

3.5 Design and performance. Ballasts furnished under this specification shall conform to the design and performance requirements of ANSI Standard C82.1. Refer to 4.4.2 for compliance testing.

3.5.1 Power factor. Ballasts furnished under this specification shall be of the high power factor type, as defined in the above ANSI Standard C82.1, having a power factor of 90 percent or above. Refer to 4.4.2, for compliance testing.

3.5.2 Multi-lamp operation. Multi-lamp ballasts shall operate lamps out of phase with each other, i.e., a two-lamp ballast shall operate lamps out of phase with each other; a three-lamp ballast shall operate one lamp out of phase with the other two, or all lamps out of phase with respect to each other; a four-lamp ballast shall operate two lamps out of phase with the other two lamps, etc. Refer to 4.4.2 for compliance testing.

3.5.3 Electromagnetic interference - free requirements. When specified (see 6.1), ballasts shall be designed for suppression of electromagnetic interference such that requirements of MIL-STD-461 are complied with. Testing for electromagnetic interference shall be in accordance with 4.5.

3.6 Marking. Each ballast exterior shall be permanently and legibly marked with the following information: Supply (input) voltage and frequency, the input current, manufacturer, manufacturer's catalog or part number, ballast type designation (preheat-starting, instant starting, rapid starting, or cold cathode), and the number, type, and watts or current of fluorescent lamps that ballast will operate. Refer to 4.4.1 for visual examination.

3.7 "Class P" designation. "Class P" ballasts shall be ballasts that have been certified by the Underwriters' Laboratories Inc., or an independent testing laboratory, acceptable to the Government as "Class P" and shall be so marked. Refer to 4.4.1 for visual examination.

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3.8 Workmanship and design. Ballasts shall be free of workmanship and design defects that may impair serviceability or appearance and workmanship and design defects that may pose a hazard to user.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.1), the supplier is responsible for the performance of all inspection requirements as specified herein and shall bear the expense of performing same. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.

4.2.1 Lot acceptance inspection. Lot acceptance inspection shall consist of a visual examination and performance testing of the end production item as outlined in 4.4.1 and 4.4.2, respectively. Sampling and acceptance levels for visual examination and performance testing shall be in accordance with 4.3.2 and 4.3.3, respectively.

4.2.2 Inspection of preparation for delivery. An inspection of preparation for delivery shall be performed in accordance with section 5 of this specification. Sampling and acceptance levels for inspection shall be in accordance with 4.6.

4.3 Sampling and acceptance levels for lot acceptance inspection.

4.3.1 Definition of lot. Lot as used herein shall be the collection of units of product from which samples are to be drawn and inspected to determine compliance with specification levels. A ballast lot shall consist of units of production of a specific Class, type, voltage rating, frequency, and capacity (see 1.2).

4.3.2 Sampling and acceptance level for visual examination. Sampling and acceptance level shall be in accordance MIL-STD-105. The Inspection Level shall be II and the Acceptable Quality Level shall be 2.5, expressed in terms of percent defective.

4.3.3 Sampling and acceptance level for performance testing. Sampling and acceptance level shall be in accordance with MIL-STD-105. The Inspection Level shall be S-3 and the Acceptable Quality Level shall be 2.5, expressed in terms of percent defective.

4.4 Lot acceptance visual examination and performance testing.

4.4.1 Visual examination. Ballast shall be visually examined to determine compliance with material requirements of paragraph 3.3, dimensional requirements of paragraph 3.4, marking requirements of paragraphs 3.6 and 3.7 of specification and of section 4 of ANSI C82.1 and workmanship requirments of paragraph 3.8. Ballasts that fail to comply with any of the above requirements shall be considered defective.

4.4.2 Performance test and design examination. In order to determine compliance with 3.5, ballasts shall be tested in accordance with section 5 of ANSI Standard C82.1 and examined in accordance with sections 6, 7, 8 and 9 of the ANSI C82.1 Standard. Compliance with 3.5.1 and 3.5.2 shall also be determined during ANSI C82.1 testing. Ballasts that fail to comply with any of the above requirements shall be considered defective.

4.5 Electromagnetic interference test. When specified (see 3.5.3 and 6.1), electromagnetic interference suppression tests shall be conducted in accordance with MIL-STD-461, and 462. Refer to MIL-STD-463, as required, for definitions and systems of units used in electromagnetic interference technology. The sample for this test shall be one complete ballast of the Class, type, voltage rating, frequency, and capacity specified in the Invitation for Bids. Test shall be conducted by an independent laboratory acceptable to the Government. Additionally, this test need be conducted only per the first lot of ballasts, provided there is no subsequent change in ballast design. Failure to meet electromagnetic interference suppression requirements (see 3.5.3) shall be cause for rejection.

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4.6 Inspection of preparation for delivery. An inspection shall be made to determine that packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table I. The sample unit shall be one shipping container packed as specified per 5.2, prepared for delivery and selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Any component missing or damaged. Packaging or packing material not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flaps. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A. Each ballast shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.2 Level B. Each ballast shall be packaged in accordance with MIL-P-116, Method III. The unit container shall conform to PPP-B-636, class weather-resistant. Closure and sealing shall be in accordance with the box specification, Method V.

5.1.2.1 Civil agencies. Each ballast shall be packaged in a box conforming to PPP-B-636, Class Domestic. Closure shall be in accordance with the box specification.

5.1.3 Intermediate packaging. When specified (see 6.1), ballasts of like descriptions shall be intermediate packaged in a box conforming to PPP-B-636, Class Domestic. The gross weight shall not exceed 40 pounds. Closure shall be in accordance with the appendix to the box specification.

5.1.4 Level C. The ballast shall be packaged in a manner that will give adequate protection against deterioration and damage during shipment.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. The ballasts shall be packed in close fitting boxes conforming to PPP-B-601, overseas type; or PPP-B-636, V2s. The contents shall be blocked and braced to prevent movement. The size or weight of the boxes shall not exceed the limitation of the box specification.

5.2.2 Level B.

5.2.2.1 Military agencies. The ballasts shall be packed as specified for Level A except the boxes shall conform to PPP-B-601, Domestic type or PPP-B-636, V3c or V3s. When specified (see 6.1) fiberboard boxes conforming to PPP-B-636 shall be Class Domestic.

5.2.2.2 Civil agencies. Ten (10) packaged ballasts shall be packed in a box conforming to PPP-B-636, class domestic. Closure shall be in accordance with the appendix to PPP-B-636.

5.2.3 Level C. The ballasts shall be packaged in a manner which will insure arrival at destination in satisfactory condition. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 Military agencies. In addition to any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class, type, voltage rating, frequency, and capacity of ballast (see 1.2 and 3.7).
(c) If other than brick type ballast is required (see 3.4).
(d) Dimensions of ballast if ballast required is of a capacity other than two-40 watt or two-100 watt lamp capacity or of a shape other than "brick type" (see 3.4).
(e) If electromagnetic interference suppression is required (see 3.5.3).
(f) Responsibility for inspection, if other than indicated (see 4.1).
(g) Packaging and packing levels required (see 5.1 and 5.2).
(h) When intermediate packaging is required (see 5.1.3).
(i) When domestic fiberboard boxes are to be used (see 5.2.2.1).

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing activity:

GSA-FSS

Review activity:

DSA-GS

User activities:

Army - EL
Navy - MC

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 20 cents each.

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